

BLUE-GREEN vs CANARY DEPLOYMENTS

Two Powerful Strategies. Different Approaches. Same Goal – **Safer Releases.**

• BLUE-GREEN DEPLOYMENT



- ✓ Deploy new version to Green environment
- ✓ Validate everything
- ✓ Switch all traffic from Blue to Green
- ✓ Fast rollback
- ✓ Simple & predictable cutover
- ✗ Requires additional infrastructure
- ✗ A sudden 100% traffic switch can expose issues

• CANARY DEPLOYMENT



VS

- ✓ Lower blast radius
- ✓ Real production validation
- ✓ Gradual rollout
- ✓ Excellent for high-traffic systems
- ✗ More complex traffic management
- ✗ Requires strong observability
- ✗ DB/Schema compatibility becomes critical

WHEN TO USE WHAT?

CHOOSE BLUE-GREEN WHEN

- You need a fast, predictable cutover
- Easy rollback is a priority
- Major releases / platform upgrades
- You can manage dual environments

CHOOSE CANARY WHEN

- You want to validate with real users
- Minimize blast radius is critical
- High-traffic, customer-facing systems
- You have strong monitoring & alerting

REAL WORLD IMPACT – COMPANIES & RESULTS

CRED

AWS RDS
Blue/Green Deployment

2 MINUTES

Average switchover time

85%

Reduction in operational effort
Across 120+ production DB clusters

ada

Blue/Green + Canary
on Amazon EKS

5 MONTHS → 5 DAYS

Kubernetes upgrades reduced from up to 5 months to 5 days

70%

Increase in deployment velocity

15%

Reduction in costs

waze

Canary Releases
(Spinnaker)

~25%

Of service incidents prevented by canary releases
By exposing changes to a small % of production traffic first

Capgemini

Global Telecom Company
(Azure Blue-Green)

95%

Reduction in deployment time

1–3 HOURS → 5–10 MINUTES

Rollback time also reduced to 5–10 minutes

wipro

Indian BFSI Client
(Blue-Green + CI/CD)

7 DAYS ↓ < 30 MINUTES

UAT delivery reduced from 7 days to < 30 mins

50%

Reduction in deployment failure rate

Infosys

Logistics Company
(Kubernetes + Blue-Green)

ZERO

Deployment downtime

50%

Reduction in effort for automated build & deployment

70%

Reduction in performance testing effort

OVERALL BUSINESS BENEFITS



Faster Time to Market



Reduced Downtime



Lower Incident Impact



Faster Rollback



Better Engineering Productivity



Higher Customer Trust



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SCAN TO CONNECT



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Let's connect and build something impactful!

💡 THE BOTTOM LINE

Deployment strategy is not just a DevOps decision.

It's an Architecture Decision.

BLUE-GREEN for speed & simplicity.

CANARY for safety & validation.

The right choice depends on your risk, architecture, and business impact.

#SoftwareArchitecture #DevOps #Microservices #Kubernetes #AKS #CloudArchitecture #DotNet #PrincipalArchitect